

Autodesk® Scaleform®

HUD

AAA

C++

Scaleform 4.3 HUD
Flash UI

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2.0

2010 7 30

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Autodesk® Scaleform® 4.3

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1 はじめに

Scaleform HUD

AAA

Direct Access API

Scaleform®
UI

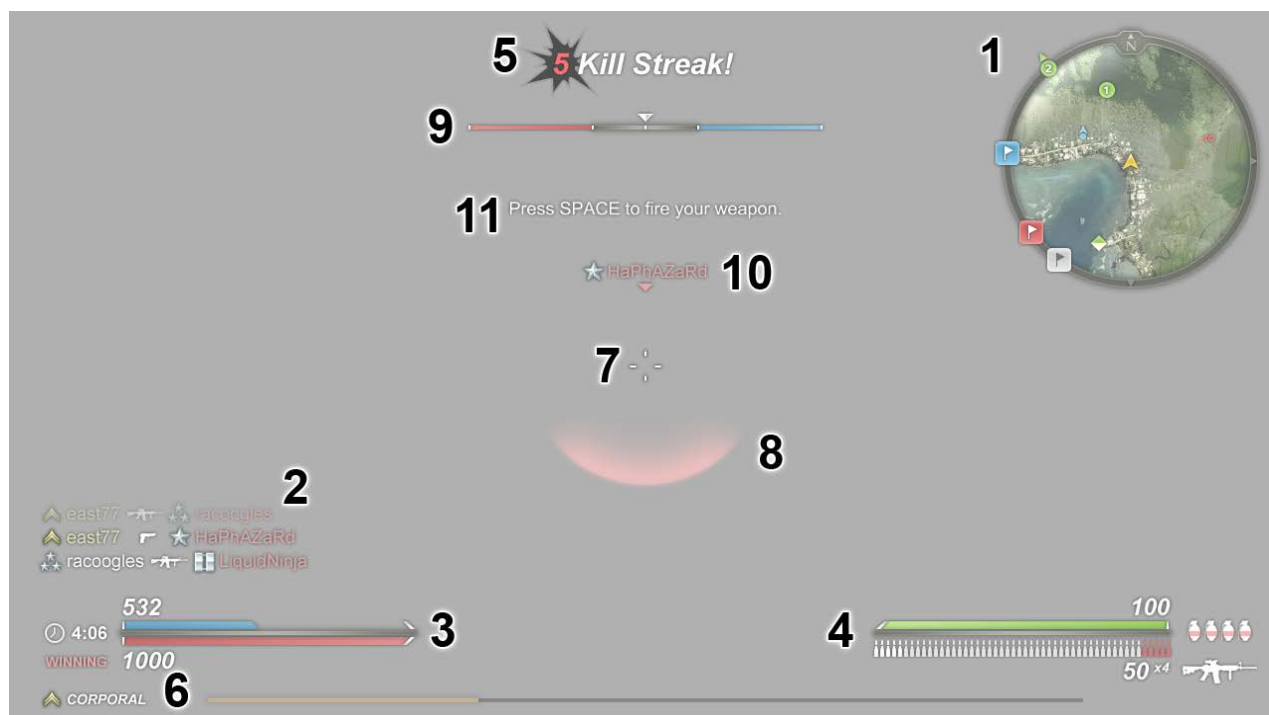


図 1: HUD の概観

HUD

UI

1. minimap
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.

FPS

UI

Scaleform

2 概要

2.1 ファイルの位置とビルドの注記

- *Apps\Kits\HUD* - HUD C++
- *Bin\Data\AS2 or AS3\Kits\HUD* - Flash ActionScript
(AS2 ActionScript2/Flash8 AS3
ActionScript3/Flash10
- *Projects\Win32 \{Msvc80 または Msvc90}\Kits\HUD* - Microsoft Windows
Visual Studio 2005/2008
- *Projects\Xbox360\{Msvc80 または Msvc90}\Kits\HUD* - Xbox 360 Visual
Studio 2005/2008
- *Projects\Common\HudKit.mk* - HUD PS3 makefile

Windows HUDKit.exe *Bin\Kits\HUD*
Scaleform SDK

Windows *Projects\Win32\Msvc80\Kits*
Msvc90\Kits Scaleform 4.3 Kits.sln

Bin\Data\AS2\Kits\HUD or Bin\Data\AS3\Kits\HUD

Xbox 360 *Projects\Xbox360\{Msvc80*
Msvc90}\Kits\HUD Scaleform 4.3 Kits.sln
Bin\Data\AS2 or AS3\Kits\HUD
Xbox360

PS3 make Scaleform
HUD

PS3 SN
Bin\PS3

o app_home/ Directory: *{Local Scaleform Directory}\Bin\Data\AS2 or AS3\Kits\HUD*

2.2 デモの使い方

HUD

2

minimap

1

500

3

1

15

×1

×1

×4

2.3 制御方法

2.3.1 キーボード (Windows)

W、A、S、D	W S A D
スペース	
R	
G	
ESC	
B	AI
1、2	• 1 • 2

2.3.2 コントローラ (Xbox/PS3)

D パッド 左ジョイスティック	W S A D
右トリガー	
X / □	
左トリガー	
B / ○	
Y / △	

3 アーキテクチャ

HUD Flash UI HUD
HUD C++
C++

3.1 C++

HUD MVC HUD C++ Flash UI
Fx
HUD HUD HUD HUD
HUD HUD
Scaleform Direct Access API
Scaleform
C++ *Apps\Kits\HUD*

3.1.1 デモ

- **HUDKitDemo.cpp** – Scaleform Player C++ SWF
- **HUDKitDemoConfig.** – FxPlayerConfig.h HUDKitDemo
Windows

3.1.2 HUD / Minimap ビューコア

- **FxHUDKit.h** – HUD HUD
- **FxHUDView.h/.cpp** – HUD
- **FxMinimap.h** – minimap
- **FxMinimapView.h/.cpp** – minimap

3.1.3 ゲームシミュレーション

- **HUDAdapter.h/.cpp** – HUD
- **HUDEntityController.h/.cpp** – AI
- **HUDSimulation.h/.cpp** – UI

3.2 Flash

HUD Flash
FLA **HUDKit.fla** **Minimap.fla** 2
Scaleform HUD UI
FLA UI

Scaleform Flash C++ ActionScript Flash
HUD HUD Classic Tweens
Flash C++
GfX::Value::GotoAndPlay()

3.2.1 HUDKit.fla

Bin\Data\AS2 or AS3\Kits\HUD HUDKit.fla
SWF HUDKitDemo
HUDKit.fla minimap Flash UI

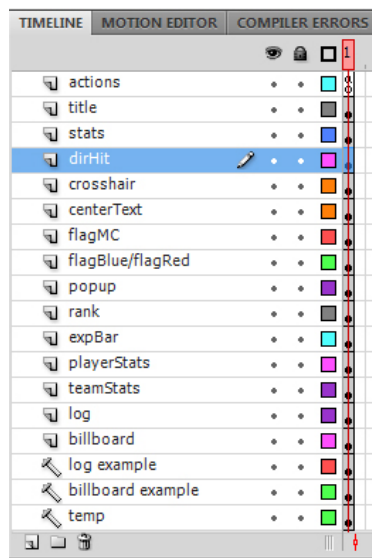


図 2: HUDKit.fla のレイヤー

HUDKitDemo

Frame 1

External Interface

HUDKit MovieClip

```
ExternalInterface.call("registerHUDView", this);
```

MovieClip

C++ C++ HUDView

MovieClip

HUD View

Scaleform C++

3.2.2 Minimap.fla

図 3: Minimap のシンボル

Minimap.fla Minimap

Minimap
3D

minimap minimap
5

-
-
-
-

C++ Stage Direct Access API MovieClip Direct Access API

Minimap

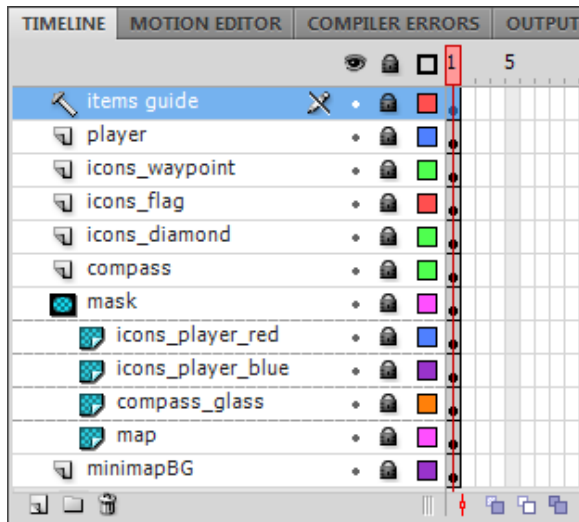


図 4: 「minimap」シンボルのレイヤー構造

- **items guide:**
SWF
- **player:**
- **icons_waypoint:**
- **icons_flag:**
- **icons_diamond:**
- **compass:**
- **mask:**
 - **icons_player_red:**
 - **icons_player_blue:**
 - **compass_glass:**
 - **map:**

C++

- **minimapBG:**

C++

Minimap com.scaleform.Minimap
AS3\Kits\HUD \com\scaleform
 HUDKitDemo Minimap
minimap

Bin\Data\AS2 or

4 HUD ビュー

C++ HUD View

HUDKit.swf

- FxHUDView – HUD HUDKit.swf MovieClip
- BillboardCache – minimap
- FxHUDLog – HUD FxHUDMessages
- FxHUDMessage – MovieClip

FxHUDView

MovieClip

MovieClip

MC

FxHUDView

UpdateView()
MovieClip

MovieClip

HUDKit.fla

UI

FxHUDView::UpdateView()

UpdateTeamStats() UpdatePlayerEXP() Update
PlayerStats() UpdateEvents() UpdateTutorial() UpdateBillboards()
UI

MovieClip
MovieClip

UpdateView()

MovieClip

Scaleform Direct Access API

Flash

GFx::Value

HUD View

Flash

GFx::Value::SetDisplayInfo()

GFx::Value::GetDisplayInfo()

GFx::Value::SetText()

MovieClip

4.1 ラウンドとの統計とスコアボード

HUDKit.fla	Scene 1	teamStats		teamStats MovieClip
			MovieClip	textField redWinning
blueWinning	2	textField		scoreRed scoreBlue
	teamRed	teamBlue		textField roundTime

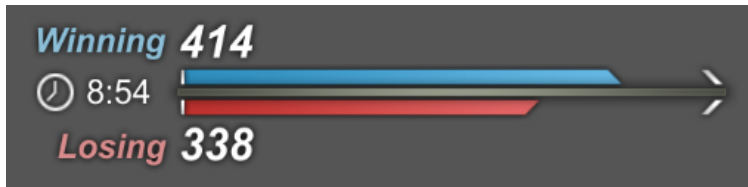


図 5: teamStats シンボル

teamStats	FxHUDView::UpdateTeamStats()	teamStats
MovieClip		

```
void FxHUDView::UpdateTeamStats(FxHUDEnvironment *penv)
{
    // Retrieve the scores from the simulation.
    unsigned scoreBlue = unsigned(penv->GetHUDBlueTeamScore());
    unsigned scoreRed = unsigned(penv->GetHUDRedTeamScore());

    String text;
    Value tf;
    Value::DisplayInfo info;

    // We won't update any element who has not changed since the last update
    // to avoid unnecessary updates. To do so, we compare the previous
    // simulation State to the latest simulation data. If the two are
    // different, update the UI element in the HUD View.
    if (State.ScoreRed != scoreRed)
    {
        Format(text, "{0}", scoreRed);
        ScoreRedMC.SetText(text); // Update the red team score text.

        info.SetScale((scoreRed / (Double)MaxScore) * 100, 100);
        TeamRedMC.SetDisplayInfo(info); // Update and scale the red team
                                         score bar movieClip.
    }

    if (State.ScoreBlue != scoreBlue)
    {
        Format(text, "{0}", scoreBlue);
        ScoreBlueMC.SetText(text); // Update the blue team score text.

        info.SetScale((scoreBlue / (Double)MaxScore) * 100, 100);
        TeamBlueMC.SetDisplayInfo(info); // Update and scale the blue team

```

```

        score bar movieClip.
    }

    if (State.ScoreRed != scoreRed || State.ScoreBlue != scoreBlue)
    {
        // Update the "Winning" and "Losing" text fields for both teams
        // if the score has changed.
        if (scoreBlue > scoreRed)
        {
            RedWinningMC.SetText(LosingText);
            BlueWinningMC.SetText(WinningText);
        }
        else
        {
            RedWinningMC.SetText(WinningText);
            BlueWinningMC.SetText(LosingText);
        }
    }

    // Update the roundtime clock
    float secondsLeft = penv->GetSecondsLeftInRound();
    unsigned sec = ((unsigned)secondsLeft % 60);
    if (sec != State.Sec)
    {
        unsigned min = ((unsigned)secondsLeft / 60);
        String timeLeft;
        Format(timeLeft, "{0}:{1:0.2}", min, sec);
        RoundTimeMC.SetText(timeLeft);
        State.Sec = sec;
    }
}

```

Scaleform::String		GfX::Value::SetText()	scoreRed
scoreBlue	textField	teamRed	teamBlue
GfX::Value::SetDisplayInfo()		X	0
		100	
blueWinning/redWinning	textField		2
	GfX::Value::SetText()	HUDView	
Winning	Losing		
textField		UpdateView()	
		1	GfX::Value::SetText()
roundTime	textField	String	

4.2 プレーヤー統計とアモ

HUDKit.fla Scene 1 playerStats healthN health reload playerStats weapon HUD

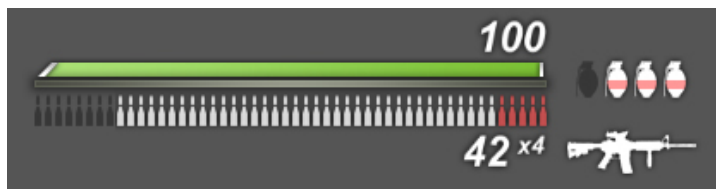


図 6: playerStats シンボル

MovieClip clipN ammoN ammo ammoBG grenade grenadeBG 6
textField clipN ammoN

ammo/healthVehicle ammo ammoBG 2

1 51 Keyframe Keyframe
Keyframe Gfx::Value::GotoAndStop() MovieClip

- ammo
- ammoBG

grenade/armor
grenade と grenadeBG 2

Frame 1 4 5 Keyframe
[Gfx::Value::GotoAndStop\(\)](#) MovieClip Frame 5 Keyframe

- grenade
- grenadeBG

playerStats

```
void FxHUDView::UpdatePlayerStats(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();

    // Load latest player info.
```

```

unsigned ammoInClip = pplayer->GetNumAmmoInClip();
unsigned ammoClips = pplayer->GetNumClips();
unsigned clipSize = pplayer->GetMaxAmmoInClip();
unsigned grenades = pplayer->GetNumGrenades();

unsigned ammoFrames = 51;
unsigned grenadeFrames = 5;

String text;

if(pplayer->GetHealth() != State.Health)
{
    unsigned health = unsigned(pplayer->GetHealth() * 100);
    Format(text, "{0}", health);
    HealthNMC.SetText(text); // Update the health text field.

    Value::DisplayInfo info;
    info.SetScale(Double(health), 100);
    HealthMC.SetDisplayInfo(info); // Update and scale the health bar.
}

if(ammoInClip != State.AmmoInClip)
{
    Format(text, "{0}", ammoInClip);
    AmmoNMC.SetText(text); // Update the ammo text field.

    // The ammo is divided into two parts: the white bullet icons (AmmoMC)
    // and their respective backgrounds (AmmoBGMC).

    // To update the number of bullets currently in the clip, we use
    // GotoAndStop with AmmoMC and select the frame with the proper
    // number of white bullet icons. There are 51 frames in AmmoMC
    // (defined above in ammoFrames) and the first frame displays every
    // bullet icon. To display X number of bullets, we subtract X from
    // the total number of frames in AmmoMC (51) and GotoAndStop on the
    // result.
    AmmoMC.GotoAndStop(ammoFrames - ammoInClip);
}

if(ammoClips != State.AmmoClips)
{
    Format(text, "x{0}", ammoClips);
    ClipNMC.SetText(text); // Update the remaining ammo clips text field.
}

if(grenades != State.Grenades)
{

```

```

        // Grenades are setup similar to Ammo. The first frame for the
        // Grenades movieClip shows 4 white grenade icons. The second frame
        // shows 3.

        // To display X number of grenades, we subtract X from the total
        // number of frames in GrenadeMC (5) and GotoAndStop on the result.
        // Note: GrenadeMC actually contains 10 frames but 6-10 are unused by
        // this implementation.
        GrenadeMC.GotoAndStop(grenadeFrames - grenades);
    }

    if (pplayer->GetWeaponType() != State.weaponType)
    {
        // Change the weapon icon if the player has changed weapons.
        unsigned weaponFrame = GetWeaponFrame(pplayer->GetWeaponType());
        WeaponMC.GotoAndStop(weaponFrame);

        // Update the ammo background icons based on clipSize.
        if (clipSize != State.AmmoClipSize)
            AmmoBGM.C.GotoAndStop(ammoFrames - clipSize);
    }

    // If the ammo in clip is less than 6 bullets, play the "Reload" indicator
    // movieClip.
    if (ammoInClip <= 5)
    {
        // ReloadMC will play until we GotoAndStop on Frame 1, so no need to
        // start the animation more than once.
        if (!bReloadMCVisible)
        {
            bReloadMCVisible = true;
            ReloadMC.GotoAndPlay("on"); // Will cycle back to "on" frame
                                         when it reaches the end and play.
                                         again
        }
    }

    // If the reload indicator is displayed but there are more than six bullets
    // hide it and stop the animation by playing frame 1 of the movieClip.
    else if (bReloadMCVisible)
    {
        ReloadMC.GotoAndStop("1"); // Frame 1 contains stop();
        bReloadMCVisible = false;
    }
}

```

<i>ammo</i>	Frame 1	50		Frame 51	0
			Keyframe		
	ammoMC.GotoAndStop(ammoFrames - ammoInClip)				HUD
	1				
<i>ammoBG</i>	ammoBGMC.GotoAndStop()			X	
	Keyframe			X	
		50		ammoBG.GotoAndStop(1)	50
	Frame 1			MovieClip	
	FxHUDView::UpdateView()				

4.3 フラグキャプチャインジケータ

HUDKit.fla Scene 1 flag *flagMC*

CaptureState *flag* *arrow* 2

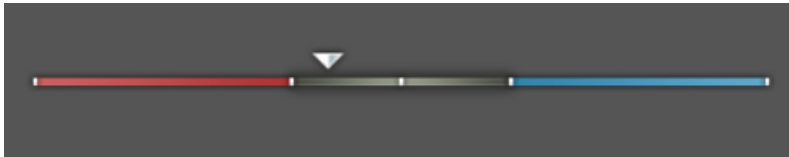


図 7: *flagMC* シンボル

flagMC *flagMC* Classic Tween Flash

Hidden Visible Fade-In FadeOut 4 5 Keyframe

GotoAndStop() Keyframe labels/numbers 1 5 "On" "Off"

FxHUDView::UpdateEvents() *flagMC*

arrow

```
// Flag capture indicators and reticule behavior.
void FxHUDView::UpdateEvents(FxHUDEnvironment *penv)
{
    Value::DisplayInfo info;

    if (penv->IsHUDPlayerCapturingObjective())
    {
        if(!bFlagMCVisible)
        {
            SetVisible(&FlagMC, true); // Set the Flag MovieClip's
                                         visibility to true.
            FlagMC.GotoAndPlay("on"); // Play the fade-in animation once.
            bFlagMCVisible = true;
        }

        // Shift the x-coordinate of the Flag Arrow to show the capture state
        // of the flag the player is currently capturing.
        info.SetX(penv->GetHUDCaptureObjectiveState() * 177);
        FlagArrowMC.SetDisplayInfo(info);
        info.Clear();
    }
}
```



```

    }
    else if (bFlagMCVisible & !penv->IsHUDPlayerCapturingObjective())
    {
        // If the flag indicator is still visible and the player is
        // no longer capturing an objective, play the fade-out animation
        FlagMC.GotoAndPlay("off");
        bFlagMCVisible = false;
    }
}

flagMC.GotoAndPlay("on")
Visible
Frame 1
ActionScript
MovieClip
Frame 1
C++
Frame 1
flagMC
flagMC.GotoAndPlay("off")
Frame 11 20
Flash
Frame 1
Frame 1
stop();
Alpha 0
flagMC
Frame 20

arrow GfX::Value::SetDisplayInfo\(\)
CaptureState
X
GfX::Value::DisplayInfo

```

HUDKit fla Scene 1 crosshair
reticule

```
reticule
left  right  bottom  top    4
```

PlayerFire



図 8: レティクルシンボル

	FxHUDView::UpdateEvents()	FxHUDView::OnEvent()
2	ReticuleHeat	ReticuleFiringOffset
FxHUDEvent		PlayerFire

```
// If the player is firing his/her weapon, update the reticule's heat.
// The reticule elements are updated in FxHUDView::UpdateEvents() based on
// the ReticuleHeat and the ReticuleFiringOffset.
```

```
case FxHUDEvent::EVT_PlayerFire:
{
    if (ReticleHeat < 8)
        ReticleHeat += 2.5f;

    ReticleFiringOffset = 2;
}
```

FxHUDView::UpdateEvents()	2		<i>top</i>	<i>bottom</i>	<i>left</i>	<i>right</i>		
MovieClip	MovieClip		SetDisplayInfo()		MovieClip	X		Y
		6	-6	0,0		X		Y
	SetDisplayInfo()		<u>Gfx::Value::DisplayInfo</u>					
	DisplayInfo							
Gfx::Value::SetDisplayInfo()			DisplayInfo.Clear()					
DisplayInfo								

```
// Shift the XY-coordinates of the reticule elements based on the
// firing duration and rate of fire.
```

```
if (ReticleHeat > 0 || ReticleFiringOffset > 0){
    if (ReticleHeat > 1.0f)
```

```

        ReticleHeat -= .02f;

ReticleFiringOffset -= .02f;
if (ReticleFiringOffset < 0)
    ReticleFiringOffset = 0;

info.SetY((-6) - (ReticleFiringOffset + ReticleHeat));
ReticuleMC_Top.SetDisplayInfo(info);
info.Clear();

info.SetX((6) + (ReticleFiringOffset + ReticleHeat));
ReticuleMC_Right.SetDisplayInfo(info);
info.Clear();

info.SetX((-6) - (ReticleFiringOffset + ReticleHeat));
ReticuleMC_Left.SetDisplayInfo(info);
info.Clear();

info.SetY((6) + (ReticleFiringOffset + ReticleHeat));
ReticuleMC_Bottom.SetDisplayInfo(info);
info.Clear();
}

```

4.5 方向性ヒットインジケータ



- a)
- b)

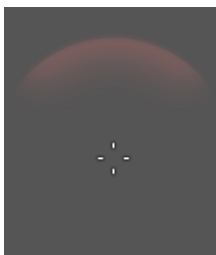
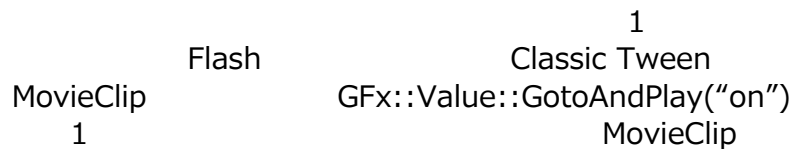


図 9: dirHit シンボル



```

void FxHUDView::OnEvent( FxHUDEvent* pevent )
{
    switch ( pevent->GetType() )
    {
        // If the Damage Event (Player was hit) is fired, show the
        // appropriate directional hit indicator.
        // We can use GotoAndPlay("on") to play the fade-in animation. It
        // will fade out on its own after ~1 second (this animation is setup
        // on the Flash timeline)
        case FxHUDEvent::EVT_Damage:
        {
            FxHUDDamageEvent* peventDamage = (FxHUDDamageEvent*)pevent;
            float dir = peventDamage->GetDirection();
            if (dir > 0)
            {
                if (dir > 90)
                {

```

```

        if (dir > 135)
            DirMC_B.GotoAndPlay("on");
        else
            DirMC_BR.GotoAndPlay("on");
    }
    else
    {
        if (dir > 45)
            DirMC_TR.GotoAndPlay("on");
        else
            DirMC_T.GotoAndPlay("on");
    }
}
else
{
    if (dir < -90)
    {
        if (dir < -135)
            DirMC_B.GotoAndPlay("on");
        else
            DirMC_BL.GotoAndPlay("on");
    }
    else
    {
        if (dir < -45)
            DirMC_L.GotoAndPlay("on");
        else
            DirMC_TL.GotoAndPlay("on");
    }
}
break;
}
}

```

4.6 ランクと経験値の表示

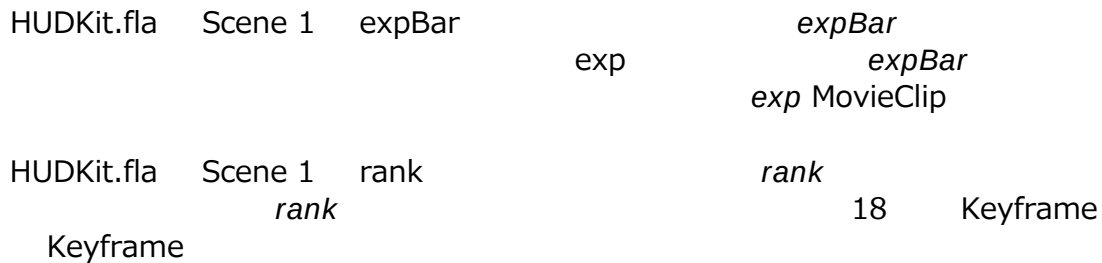


図 10: expBar と rank シンボル

```

void FxHUDView::UpdatePlayerEXP(FxHUDEnvironment *penv)
{
    FxHUDPlayer* pplayer = penv->GetHUDPlayer();
    if (pplayer->GetLevelXP() != State.Exp)
    {
        Value::DisplayInfo info;
        // Update the rank icon. Each rank icon is on a different frame of
        // the rank movieClip.
        RankMC.GotoAndStop(UInt(pplayer->GetRank() + 1));
        info.SetScale(pplayer->GetLevelXP() * 100, 100); // Scale the EXP
                                                         bar movieClip.
        ExpBarMC.SetDisplayInfo(info);
    }
}
  
```



4.7 テキスト通知

HUDKit.fla Scene 1 centerText

centerTextMC
textField *centerText*

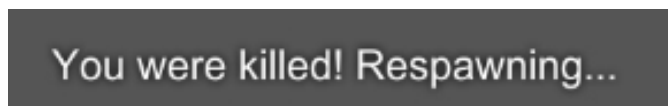
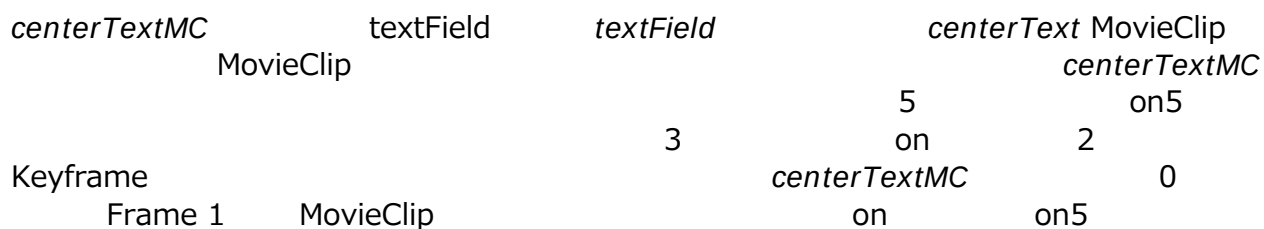


図 11: *centerTextMC* シンボル



```
TextFieldMC.SetText(RespawnText); // Set the text field of the Center Text.
CenterTextMC.GotoAndPlay("on"); // Play the fade-in animation, it will fade on
its own after ~3 sec.
```

centerTextMC *textField* MovieClip GotoAndPlay("on")
MovieClip Classic Tween 3

4.8 ポップアップと通知

HUDKit.fla Scene 1 popup popup 3

C++ 3 5



図 12: popup シンボル

Popup
Classic Tweens popup
popupNumber popupText popupBG 4

popup actions
popupBG
popupText popupNumber

textField

on Keyframe GotoAndPlay()

0 Frame 1

```
// If a KillStreak event is fired, display the kill streak pop-up.
case FxHUDEvent::EVT_KillStreak:
{
    FxHUDKillStreakEvent* peventKS = (FxHUDKillStreakEvent*)pevent;

    // Format the number of kills
    String text;
    Format(text, "{0}", peventKS->GetSrcKillStreak());
    PTextFieldMC.SetText(text); // Set the text field of the PopUp.

    // Play the fade-in animation, it will fade on its own after ~4 sec.
    PopupMC.GotoAndPlay("on");
}
break;
```

KillStreak FxHUDEvent

textField PTextFieldMC C++ GotoAndPlay("on") popupText

popup

4.9 メッセージとイベントログ

HUDKit.fla Scene 1 log

log

HUD

3



図 13: log と logMessage シンボル

logMessage

3

logMessage

C++

htmlText

textField

FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()

LogMessage MovieClip

MovieClip

Log MovieClip

FxHUDMessage::Movieclip* FxHUDLog::GetUnusedMessageMovieclip()

{

if (MessageMCs.IsEmpty())

{

// Request a new line in the SWF

Value logline;

LogMC.AttachMovie(&logline, "LogMessage", "logMessage"+NumMessages);

FxHUDMessage::Movieclip* pmc = new FxHUDMessage::Movieclip(logline);

MessageMCs.PushBack(pmc); // Add new Message MC to list of unused
message movieClips.

}

FxHUDMessage::Movieclip* pmc = MessageMCs.GetFirst(); // Use an unused
Message movieClip.

pmc->SetVisible(true);

MessageMCs.Remove(pmc);

return pmc;

```
}
```

htmlText C++ C++ Level Up
You are now a [Rank Icon] [Rank Name]

```
void     FxHUDLog::Process(FxHUDEvent *pevent)
{
    String msg;
    switch (pevent->GetType())
    {
        case FxHUDEvent::EVT_LevelUp:
        {
            FxHUDLevelUpEvent* e = (FxHUDLevelUpEvent*)pevent;
            Format(msg, "You are now a <img src='rank{1}'/> {0}!",
                e->GetNewRankName(), e->GetNewRank());
        }
        break;
    }
}

FxHUDMessage::Movieclip* pmc = GetUnusedMessageMovieclip();
FxHUDMessage* m = new FxHUDMessage(msg, pmc); // 3 seconds
Log.PushBack(m);
NumMessages++;
```

HUD Update

```
void     FxHUDLog::Update()
{
    SF_ASSERT(LogMC.IsDisplayObject());

    Double rowHeight = 30.f;

    // Tick the message lifetimes
    Double yoff = 0;
    FxHUDMessage* data = Log.GetFirst();
    while (!Log.IsNull(data))
    {
        FxHUDMessage* next = Log.GetNext(data);
        if (data->IsAlive())
        {
            // Layout mgmt and animation
            Value::DisplayInfo info;
```

```

        info.SetY(yoff);
        data->GetMovieclip()->GetRef().SetDisplayInfo(info);
    }
    data = next;
    yoff += rowHeight;
}

// Layout management and animation
Value::DisplayInfo info;
info.SetY(LogOriginalY - (NumMessages * rowHeight));
LogMC.SetDisplayInfo(info);

// Remove dead messages
data = Log.GetFirst();
while (!Log.IsNull(data))
{
    FxHUDMessage* next = Log.GetNext(data);
    if (!data->IsAlive())
    {
        Log.Remove(data);
        NumMessages--;

        FxHUDMessage::Movieclip* pmc = data->GetMovieclip();
        pmc->SetVisible(false);
        MessageMCs.PushBack(pmc);

        delete data;
    }
    data = next;
}
}

```

Log	Alpha	0	
FxHUDMessage			Log
	FxHUDMessage	MovieClip	MessageMC
FxHUDMessage		GFx::Value::DisplayInfo.SetY()	
GFx::Value::SetDisplayInfo()		MessageMC	

4.10 ビルボード

HUDKit.fla Scene 1 billboard

billboard_container

3D

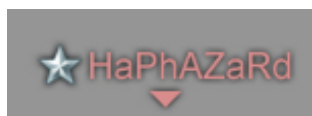


図 14: billboard_enemy シンボル

Gfx::Value::SetDisplayInfo()
 2D MovieClip UI

MovieClip billboard_friendly billboard_enemy text
 billboard_label_ friendly/enemy arrow
 billboard_label_* textField
 billboard_friendly Show Keyframe billboard_friendly Classic
 Tween Show

MovieClip C++
 FxHUDView::BillboardCache::GetUnusedBillboardMovieclip

```
if (UnusedBillboards.IsEmpty())
{
    Value::DisplayInfo info(false);

    Value billboard, temp;
    String instanceName;
    Format(instanceName, "{0}{1}", BillboardPrefix, BillboardCount++);
    CanvasMC.AttachMovie(&billboard, SymbolLinkage, instanceName);

    BillboardMC* pbb = new BillboardMC();
    pbb->Billboard = billboard;
    billboard.GetMember("label", &temp);
    temp.GetMember("textField", &temp);
    pbb->Textfield = temp;

    pbb->Billboard.SetDisplayInfo(info);
}
```

```

        UnusedBillboards.PushBack(pbb);
    }

```

```

BillboardCache      CanvasMC
                    billboard_enemy/billboard_friendly
CanvasMC.AttachMovie()      MovieClip
                                BillboardCache
                                MovieClip

```

```

Billboard      UpdateBillboards()  BeginProcessing()  EndProcessing()
GetUnusedBillboardMovieclip()      BeginProcessing()
EndProcessing()

```

```

    UpdateBillboards()

```

```

// Process friendlies
// Friendly billboards will always show names

```

```

BillboardMC* pbb = NULL;

```

```

FriendlyBillboards.BeginProcessing();
for (unsigned i=0; i < friendlies.GetSize(); i++)
{
    info.Clear();
    if (FriendlyBillboards.GetUnusedBillboardMovieclip(friendlies[i].pPlayer,
                                                         &pbb))
    {
        info.SetVisible(true);
        pbb->Billboard.GotoAndPlay("show");
        // Load player info
        String title;
        Format(title,
               "<img src='rank{0}' width='20' height='20' align='baseline'
               vspace='-7' /> {1}",
               friendlies[i].pPlayer->GetRank()+1,
               friendlies[i].pPlayer->GetName());
        pbb->Textfield.SetTextHTML(title);
    }
    info.SetX(friendlies[i].X);
    info.SetY(friendlies[i].Y);
    info.SetScale(friendlies[i].Scale, friendlies[i].Scale);
    pbb->Billboard.SetDisplayInfo(info);
}

```

```
FriendlyBillboards.EndProcessing();
```

```
UpdateBillboards()
```

```
UpdateBillboards()  
GotoAndPlay("show")      MovieClip    X    Y  
GFx::Value::SetDisplayInfo()      GFx::Value::SetText()
```

```
textField    htmlText
```

```
HTML  
>GetRank()+1      img      1      src='rank{0}'  
                                friendlies[i].pPlayer-  
                                {1}      vspace  
                                friendlies[i].pPlayer->GetName()
```

5 Minimap ビュー

Minimap C++

Apps\Kits\HUD

- **FxMinimap.h**: minimap
- **FxMinimapView.h/.cpp**: minimap

5.1.1 Minimap ビュー

minimap

- **FxMinimapIconType**: ID
- **FxMinimapIcon**:
- **FxMinimapIconCache**:

MinimapIconFactoryBase

MovieClip

MovieClip

MovieClip

minimap

textField

Update()

o *PlayerIcon*:

o *FlagIcon*

o *ObjectiveIcon*

o *WaypointIcon*

- **FxMinimapView**: minimap

UpdateView()

PlayerIcon::Update()

Direct Access API

MovieClip

SetMember()

FxMinimapView.cpp 116

MovieClip

SetDisplayInfo

MovieClip

GFx::Value::SetMember

GFx::Movie::SetVariable

```
virtual void Update(FxMinimapView* pview, FxMinimapEntity* pentity,
```

```

float distSquared)

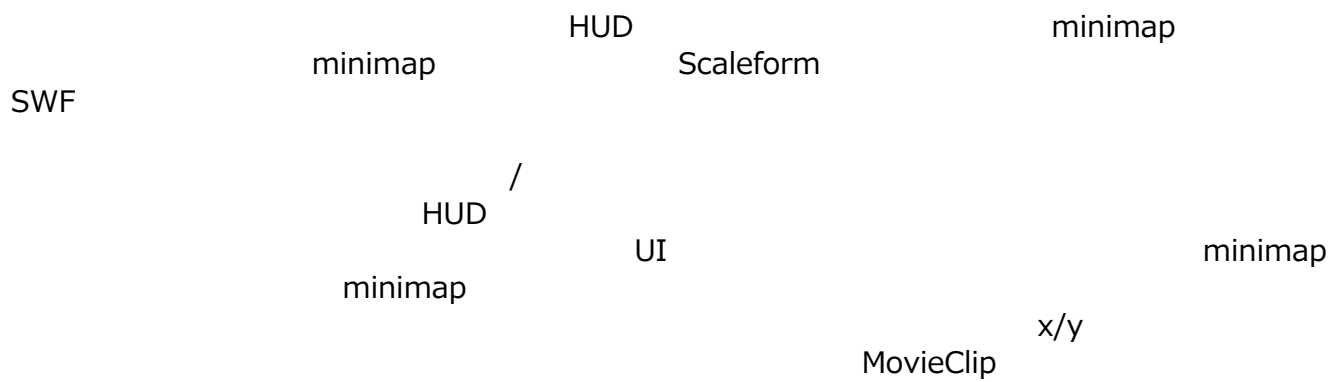
{
    SF_UNUSED(distSquared);
    bool hasIcon = (pentity->GetIconRef() != NULL);
    PointF pt = pentity->GetPhysicalPosition();
    bool showDir = pview->IsShowingPlayerDir();

    // If entity did not have an icon attached before, then wait for the
    // next update to set the state data. picon is most probably not
    // initialized (the movie needs to advance after creation)
    if (hasIcon && (State != (int)showDir))
    {
        // state 0: no arrow, state 1: show arrow
        Value frame(Double(showDir ? 2.0 : 1.0));
        MovieClip.Invoke("gotoAndStop", NULL, &frame, 1);
        State = (SInt)showDir;
    }

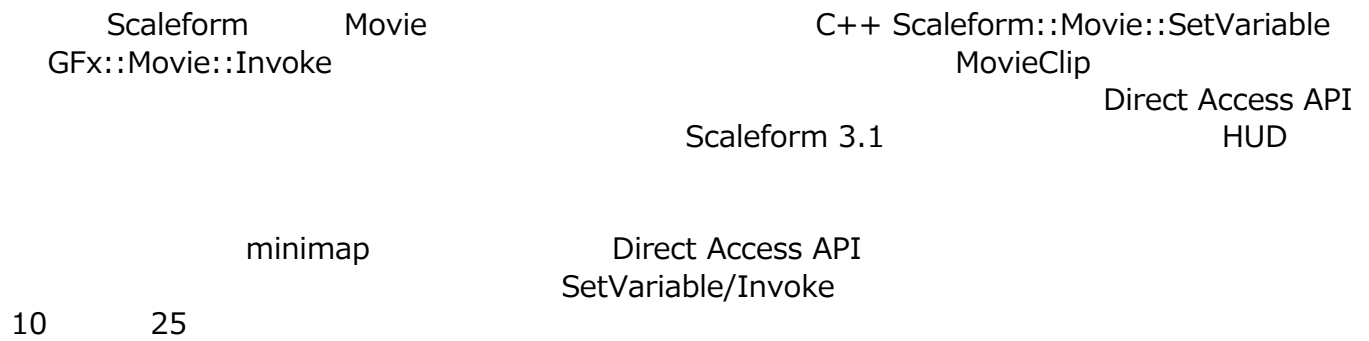
    pt = pview->GetIconTransform().Transform(pt);
    Value::DisplayInfo info;
    if (State)
    {
        info.SetRotation(pview->IsCompassLocked() ?
            (pentity->GetDirection() + 90) : (pentity->GetDirection() -
                pview->GetPlayerDirection()));
    }
    info.SetPosition(pt.x, pt.y);
    MovieClip.SetDisplayInfo(info);
}

```


6 性能解析



6.1 性能統計



6.2 メモリの分析

HUD

 GfxExport
 .GFX

1. ファイルフォーマット : GFX スタンダードエクスポート、イメージ非圧縮

全メモリ: 1858 KB

- HUDKit.gfx - 33 KB
- Minimap.gfx - 50 KB
- fonts_en.gfx - 46 KB
- gfxfontlib.gfx - 100 KB

非圧縮イメージ

- HUDKit.gfx
 - o UI - 603 KB
 - o - 72 KB
 - o - 51 KB
 - 備考: HUD 43 KB
- Minimap.gfx
 - o - 644 KB
- Map.jpg - 259 KB